# Problem 1: Finding Complexity using Counter Method

**Started on** Thursday, 6 August 2026, 10:13 AM

**State** Finished

**Completed on** Thursday, 6 August 2026, 10:59 AM

**Time taken** 46 mins 38 secs

**Marks** 1.00/1.00

**Grade** **10.00** out of 10.00 (**100**%)

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

| Input | Result |
|---|---|
| 9 | 12 |

**Answer:** (penalty regime: 0 %)

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Falling back to raw text area.

```
{
    int n;
    scanf("%d", &n);
    int i = 1;
    int s = 1;
    int counter = 2;
    while (1)
    {
        counter++;
        if (s > n)
            break;
        i++;
        counter++;
        s += i;
        counter++;
    }
    printf("%d", counter);
    return 0;
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9 | 12 | 12 | ✔ |
| ✔ | 4 | 9 | 9 | ✔ |

Passed all tests! ✔

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:** (penalty regime: 0 %)

```
1 #include <stdio.h>
2 void func(int n)
```

**Answer:** *(penalty regime: 0 %)*

```c
#include <stdio.h>
void func(int n)
{
    int count = 0;
    for(int i = 1; i <= n; i++)
    {
       count++;
       for(int j = 1; j <= n; j++)
       {
          count++;
          count++;
          count++;
          count++;
          break;
       }
    }
   printf("%d\n", count + 2);
}
int main() {
    int n;
    if (scanf("%d", &n) == 1) {
        func(n);
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 2 | 12 | 12 | ✔ |
| ✔ | 1000 | 5002 | 5002 | ✔ |
| ✔ | 143 | 717 | 717 | ✔ |

Passed all tests! ✔

# Problem 3: Finding Complexity using Counter Method

**Started on** Thursday, 17 September 2026, 10:08 AM

**State** Finished

**Completed on** Thursday, 17 September 2026, 10:20 AM

**Time taken** 12 mins 3 secs

**Marks** 1.00/1.00

**Grade** **10.00** out of 10.00 (**100**%)

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

```c
#include <stdio.h>

void Factor(int num)
{
    int c = 0;

    c++;

    for (int i = 1; i <= num; ++i)
    {
        c++;

        if (num % i == 0)
        {
            c++;
        }

        c++;
    }

    printf("%d", c);
}

int main()
{
    int n;
    scanf("%d", &n);

    Factor(n);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 31 | 31 | ✔ |
| ✔ | 25 | 54 | 54 | ✔ |
| ✔ | 4 | 12 | 12 | ✔ |

Passed all tests! ✔

# Problem 4: Finding Complexity using Counter Method

**Started on** Thursday, 17 September 2026, 10:14 AM

**State** Finished

**Completed on** Thursday, 17 September 2026, 10:23 AM

**Time taken** 9 mins 6 secs

**Marks** 1.00/1.00

**Grade** **10.00** out of 10.00 (**100**%)

**Question 1** | Correct | Mark 1.00 out of 1.00 | Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

```c

#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int c = 1;
    int outer = n - n / 2;

    int jcount = 0;
    int j = 1;

    while (j < n)
    {
        jcount++;
        j = j * 2;
    }

    c += outer + 1;
    c += outer * (jcount + 1);
    c += outer * jcount * (jcount + 1);
    c += outer * jcount * jcount;

    printf("%d", c);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4 | 30 | 30 | ✔ |
| ✔ | 10 | 212 | 212 | ✔ |

Passed all tests! ✔

**Correct**

Marks for this submission: 1.00/1.00.

# Problem 5: Finding Complexity using counter method

**Started on** Thursday, 17 September 2026, 10:15 AM

**State** Finished

**Completed on** Thursday, 17 September 2026, 10:24 AM

**Time taken** 8 mins 31 secs

**Marks** 1.00/1.00

**Grade** **10.00** out of 10.00 (**100**%)

**Question 1** | Correct | Mark 1.00 out of 1.00 | Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
 print(rev);
 }
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int rev = 0, remainder;
    int c = 0;

    c++;   // rev = 0

    while (n != 0)
    {
        c++;

        remainder = n % 10;
        c++;

        rev = rev * 10 + remainder;
        c++;

        n /= 10;
        c++;
    }

    c++;

    c++;

    printf("%d", c);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 11 | 11 | ✔ |
| ✔ | 1234 | 19 | 19 | ✔ |

Passed all tests! ✔